MILITARY SPECIFICATION

CONTROL UNIT SUBSYSTEM

FOR THE

BOMBING SET, ANGLE RATE AN/ASB-19(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification establishes the performance, functional, and test requirements for an airborne control unit for the AN/ASB-19(V) Angle Rate Bombing Set. The Control Unit Subsystem (CUS) will herein be referred to as the "subsystem".

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Control, Bombing Set, Angle Rate C-10122/ASB-19(V)	3.4.1
Control-Indicator ID-2085/ASB-19(V)	3.4.2

1.3 Associated Equipment - This equipment shall operate with the following associated ARBS equipment.

Equipment	Applicable Specification
ARBS Dual Mode Tracker	MIL-D-85056(AS)
ARBS Weapon Delivery Computer	MIL-W-85057(AS)

Comments or recommendations which may be of use in improving this document may be sent to the preparing activity when a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appears at the end of this document, or by letter addressed to the cognizant activity.

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2. APPLICABLE DOCUMENTS

2.1 Government Documents - Unless otherwise specified herein, the following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-T-18303	Test Procedures, Preproduction, Acceptance and Life for Aircraft Electronics Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment
MIL-C-81774	Control Panel, Aircraft, General Requirements for
MIL-A-85055(AS)	Avionics System Performance Specification for A-4M/AV-8 Bombing Set, Angle Rate AN/ASB-19(V)
MIL-D-85056(AS)	Dual Mode Tracker for Bombing Set, Angle Rate AN/ASB-19(V).
MIL-W-85057(AS)	Weapons Delivery Computer for the Bombing Set, Angle Rate AN/ASB-19(V).

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STANDARDS

Military

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704A Notice 3 11 April 1973	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781B	Reliability Tests Exponential Distribution
MIL-STD-810 Notice 4 21 September	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment and Facilities
MS-25213	Control Panel, Aircraft Equipment, Typical Installation

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

(2) Copies of this specification may be obtained upon application to Naval Air Systems Command, (Navy) Attention: Avionics Division, AIR 5493.

2.2 Non-Government Documents - The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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STANDARDS

Hughes Aircraft Company
(Code Ident 82577)

Documents

X3366000-350	Electrical Interface Document

Drawings

X3366040-500	Installation Control - Control Unit
X3366050-500	Installation Control - Data Entry Unit

2.2.1 Availability of Non-Government Documents - Non-Government documents shall be available from the contracting officer.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - Selection of parts and materials shall be in accordance with the requirements of MIL-A-85055.

3.3 Design and Construction - The subsystem shall conform to the applicable requirements of MIL-A-85055, except as otherwise specified herein.

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3.3.1 Physical Characteristics - The physical characteristics of the subsystem shall be as specified herein, and in accordance with MIL-A-85055.

3.3.1.1 Total Weight - The total weight of the subsystem, excluding cables, shall be a minimum consistent with good design and shall not exceed 9.0 pounds.

3.3.1.2 Dimensions - The dimensions shall not exceed the envelopes as specified on the following drawings:

(1) Control, Bombing Set Angle Rate 3366040-500

(2) Control-Indicator 3366050-500

3.3.1.3 Mounting - Mounting in the cockpit will be in accordance with MS-25213.

3.3.2 Reliability - The contractor shall conduct a reliability program in accordance with MIL-A-85055.

3.3.2.1 Operational Stability - The subsystem shall operate with satisfactory performance, continuously or intermittently, for a period of at least 3200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The subsystem shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor and approved by the procuring activity.

3.3.2.3 Reliability in Mean-Time-Between-Failure (MTBF) - The subsystem, including any Built-in-Test (BIT) provisions, shall have specified Θ_0 MTBF of 3200 hours when tested as specified in MIL-A-85055. The minimum acceptable Θ_1 MTBF shall be 1600 hours.

3.3.2.4 Time Totalizing Meter - The subsystem shall contain a time totalizing meter in accordance with MIL-A-85055. The time totalizing meter need not be mounted on the front panel.

3.3.3 Cabling and Connectors - Cabling and connectors shall conform to the requirements of MIL-A-85055.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

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3.3.5 Interchangeability - The subsystem shall meet the interchangeability requirements of MIL-A-85055

3.3.6 Electromagnetic Compatibility - The subsystem shall be designed to conform to the requirements of MIL-A-85055 for electromagnetic compatibility.

3.3.6.1 Electromagnetic Interference (EMI) - The subsystem shall be designed to conform to the requirements of MIL-A-85055 for electromagnetic interference.

3.3.7 Maintainability Provisions - The Maintainability Program, BIT features, construction and packaging, provisions for test points, and other maintainability parameters shall conform to MIL-A-85055.

3.3.8 Nomemclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307. Marking shall be in accordance with MIL-STD-130.

3.3.8.1 Serialization - Each shop repairable assembly (SRA) designed for normal replacement shall have a permanent serial number. Serial numbers shall be applied at the final assembly of these units. In the event a unit is scrapped, the serial number shall not be reused.

3.3.9 Standard Conditions - The conditions used to establish normal performance characteristics under standard conditions and for making laboratory bench tests shall be as specified in MIL-A-85055.

3.3.10 Service Conditions - The subsystem, while operating or nonoperating, shall withstand the natural and induced environments as specified herein, including all logical adverse combinations with repeated exposure to conditions having cumulative adverse effects, in accordance with MIL-A-85055.

3.3.10.1 Nonoperating Conditions - The subsystem shall be capable of withstanding the temperature, humidity, vibration, and shock as shown in Table I without physical damage or subsequent degradation of performance. Temperature and altitude limits for transportation and storage shall be as follows:

(1) High temperature: 85°C (185°F).

(2) Low temperature: -54°C (-65°F).

(3) Altitude: 50,000 feet above mean sea level (MSL).

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3.3.10.2 Operating Temperature-Altitude - The subsystem shall meet the performance requirements of 3.4 and 3.5 when exposed to the temperature altitude environments shown in Figure 1, as follows:

(1) The subsystem shall be capable of continuous operation at the extreme limits of Figure 1, Curve A. Continuous operation is defined as indefinite operation within the extremes of Curve A.

(2) The subsystem shall be capable of performing intermittent operations at the extreme limits of Figure 1, Curve B. Intermittent operation is defined as 45 minutes operation (any selected mode) within the limits of Curve A, followed by 30 minutes operation (any selected mode) at extreme conditions of Curve B, followed by 45 minutes operation (any mode) within limits of Curve A.

(3) The subsystem shall be capable of performing intermittent operations at extreme limits of Figure 1, Curve B, and Point 1, without part failure, but performance degradation may occur. Extreme intermittent operation of the subsystem at Point 1 is defined as 15 minutes operation (70°C) followed by operation within the limits of Curve A.

(4) The subsystem shall be capable of starting at a low temperature condition of - 40°C and perform as specified in 3.4 and 3.5 after the warmup period specified in 3.3.11.

3.3.10.3 Humidity - The subsystem shall withstand the effects of humidity in accordance with Table I.

3.3.10.4 Vibration - The subsystem shall not suffer damage or subsequently fail to provide the performance specified herein when subjected to the vibration levels in accordance with Figure 2.

3.3.10.5 Shock - The subsystem shall not suffer damage or subsequently fail to provide the performance specified herein when subjected to the level of shock as specified in Table I.

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Table I. ARBS Environmental Criteria-Factory to Target

Environment Item	Event – Condition									
	Transportation				Storage		Handling		Aircraft/Mission	
	Truck	Rail	Ship	Air	Airfield	Carrier	Airfield	Carrier	Non-Oper	Operating
Temperature High ^{o}C (^{o}F) Low ^{o}C (^{o}F)	50 (122) -30 (-22)	50 (122) -30 (-22)	38 (100) -18 (0)	60 (140) -18 (0)	70 (158) -30 (-22)	38 (100) 0 (32)	50 (122) -30 (-22)	30 (86) 0 (32)	85 (185) -54 (-65)	Figure 1 -40 (-40)
Shock ^{o}C/min	NEG	NEG	NEG	NEG	NEG	NEG	NEG	NEG	80	
Altitude feet	NEG	NEG	NEG	50,000	NEG	S/L	NEG	S/L	50,000	S/L to 45,000
Alt rate ft/min	NEG	NEG	NEG	1,000	N/A	N/A	N/A	N/A	40,000	
Vibration Sinusoidal, g(PK) Freq range, (Hz)	2 g, 5-12 Hz 0.5g, 12-100 Hz	1 g 5-500 Hz	0.7 g 5-70 Hz	0.4 g, 5-40 5 g 40-700	N/A N/A	N/A N/A	N/A N/A	N/A N/A		
Vibration Random g^2/Hz Freq range, (Hz)	See above				N/A	N/A	N/A	N/A	$0.02g^2$/Hz (Figure 2)	
Vibration Gunfire g^2/Hz Freq range (Hz)	N/A								MIL-STD-810B, Meth 519.1 CU: 0.05 g^2/Hz (Figure 2)	
Acceleration Shock (S) Load (L) G (PK), time Waveform	Transportation (in container): 25 g, 25 ms, 1/2 sine wave Loading/unloading (in container): IAW MIL-STD-810B, Method 516, Procedure II								Shock 15 g 11 ms 1/2 sine wave	Load Vert: +7 to -1g Long: ±1.5g Lat: ±1.1g
Acoustic vib, dB	140 dB 53 – 6,784 Hz									
Humidity, lb water per lb dry air	100% humidity over a temperature range from 0°C (0.004 lb water per lb dry air) to 35°C (0.031 lb water per lb dry air)									
Corrosion potential salt spray test	N/A	N/A	Severe	N/A	N/A	Severe	N/A	Severe	Severe	

NEG = NEGLIGIBLE N/A = NOT APPLICABLE

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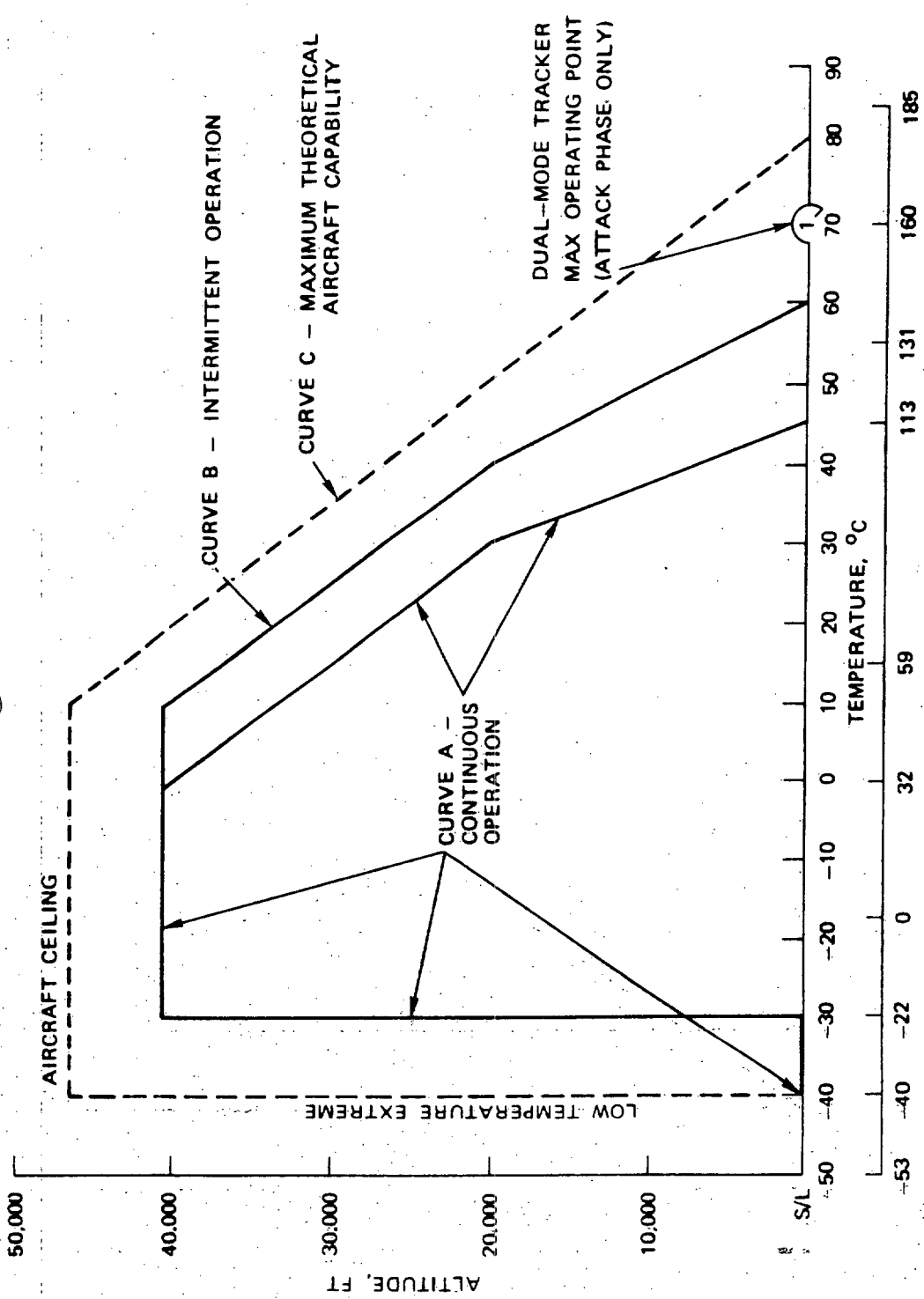


Figure 1. ARBS Equipment Operating Temperature/Altitude

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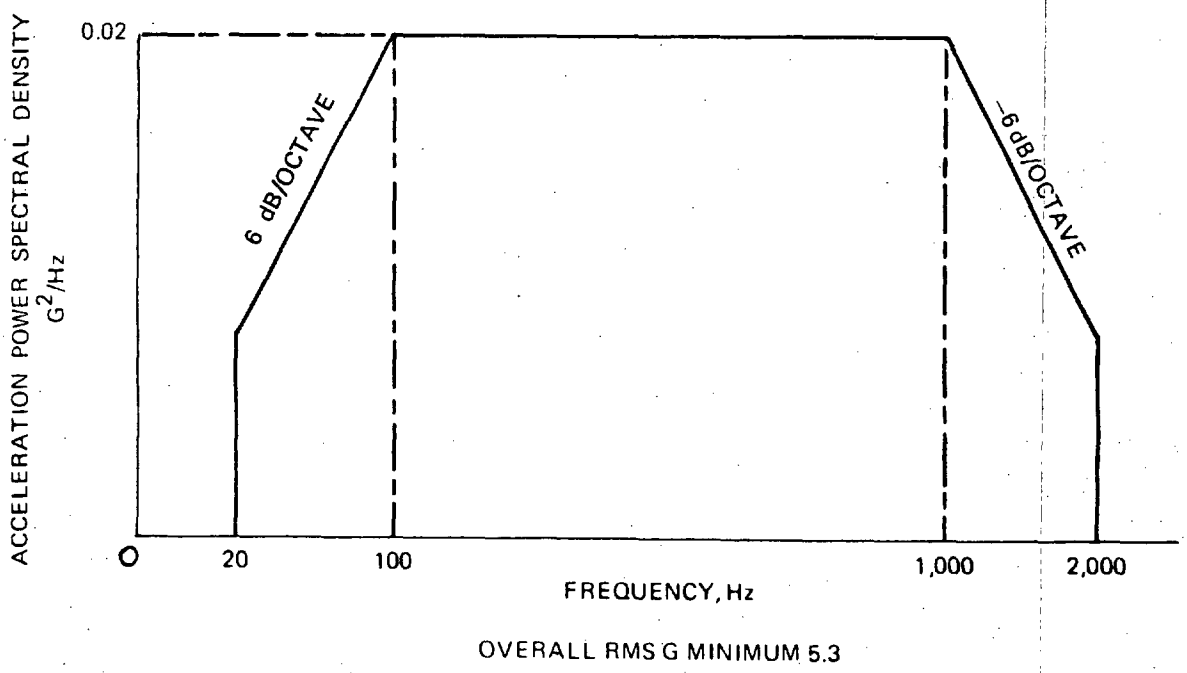


Figure 2. Random Vibration Levels.

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3.3.10.6 Gunfire - The subsystem shall be capable of withstanding, in the operating mode, vibrations induced by gunfire as defined in Table I with no physical damage or degradation of performance.

3.3.10.7 Acoustic Noise - The subsystem shall be capable of performing in accordance with 3.4 and 3.5, after being subjected to sound pressure levels as specified in Table I for periods as long as 30 minutes, and shall suffer no physical damage as a result of exposure to acoustic noise, in accordance with MIL-E-5400.

3.3.10.8 Salt Atmosphere - The subsystem shall withstand exposure to salt-sea atmosphere, in accordance with MIL-E-5400.

3.3.10.9 Sand and Dust - The subsystem shall withstand exposure to sand and dust as specified in MIL-E-5400.

3.3.10.10 Explosive Condition - The subsystem shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere, in accordance with MIL-E-5400.

3.3.10.11 Fungus - The subsystem shall withstand exposure to fungus as specified in MIL-E-5400.

3.3.11 Warm-Up Time - The time required for the subsystem to warm-up prior to operation shall be kept to a minimum and shall not exceed 30 seconds under temperature conditions of Curves A and B, Figure 1, and 5 minutes at the temperature conditions of Curve C, Figure 1.

3.3.12 Input Electrical Power

3.3.12.1 Operating Power - The subsystem shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts:

(1) AC Power (Single Phase), not applicable (NA).

(2) AC Power (Three Phase), not applicable.

(3) DC Power, 28 V, Category B, 10 W.

3.3.12.2 Lighting Power - Input power for lighting shall be as specified in the following subparagraphs.

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3.3.12.2.1 Panel - Input power for panel lighting shall be not greater than 1.8 amperes (A) at 5 volts alternating current (Vac), 400 Hertz (Hz).

3.3.12.2.2 Caution - Input power for caution lights shall be not greater than 0.2 A at 7.4 Vac, 400 Hz or 0.2 A at 28 Vac, 400 Hz.

3.3.12.2.3 Instrument - Input power for instrument lights shall be not greater than 1.8 A at 5 Vac, 400 Hz.

3.3.12.3 Degraded Performance - Degraded performance will be permitted for voltage transients of less than 0.5 second during normal electric system operation, except the performance may not be affected by short pulses of less than 50 microseconds. After transients, operation shall return to normal with no resulting damage to the subsystem.

3.3.12.4 Circuit Protection - Overload protection shall be provided for all parts and circuits which are likely to carry an overload due to any failures or poor adjustments, and shall withstand such overload without permanent damage or shall have suitable protective devices. Fuses shall not be used. Power supplies of the current limiting type shall be used to the maximum extent possible.

3.3.13 Interface Definition - Interface requirements for the subsystem shall be as defined in the following paragraphs in accordance with MIL-A-85055.

3.3.13.1 External Interfaces - The subsystem functional block diagram of Figure 3, shall interface with the associated equipment listed in 1.3 and the following:

(1) A-4M aircraft.

(2) AV-8A aircraft.

(3) WDC.

(4) DMT.

(5) Laser Spot Tracker (LST) Scan Switch.

3.3.13.1.1 A-4M Aircraft - The subsystem will physically be mounted in the cockpit of the A-4M aircraft and shall interface with the A-4M aircraft in accordance with MIL-A-85055. Electrical power supplied to the subsystem will have the characteristics as defined in MIL-STD-704 and as follows:

(1) 0 to 5 Vac for console lighting.

(2) 0 to 5 Vac for numeric display lighting.

(3) 28 Vdc day-night lighting discrete.

(4) 28 Vdc for power ON.

(5) 7.4 or 28.0 Vac caution light power.

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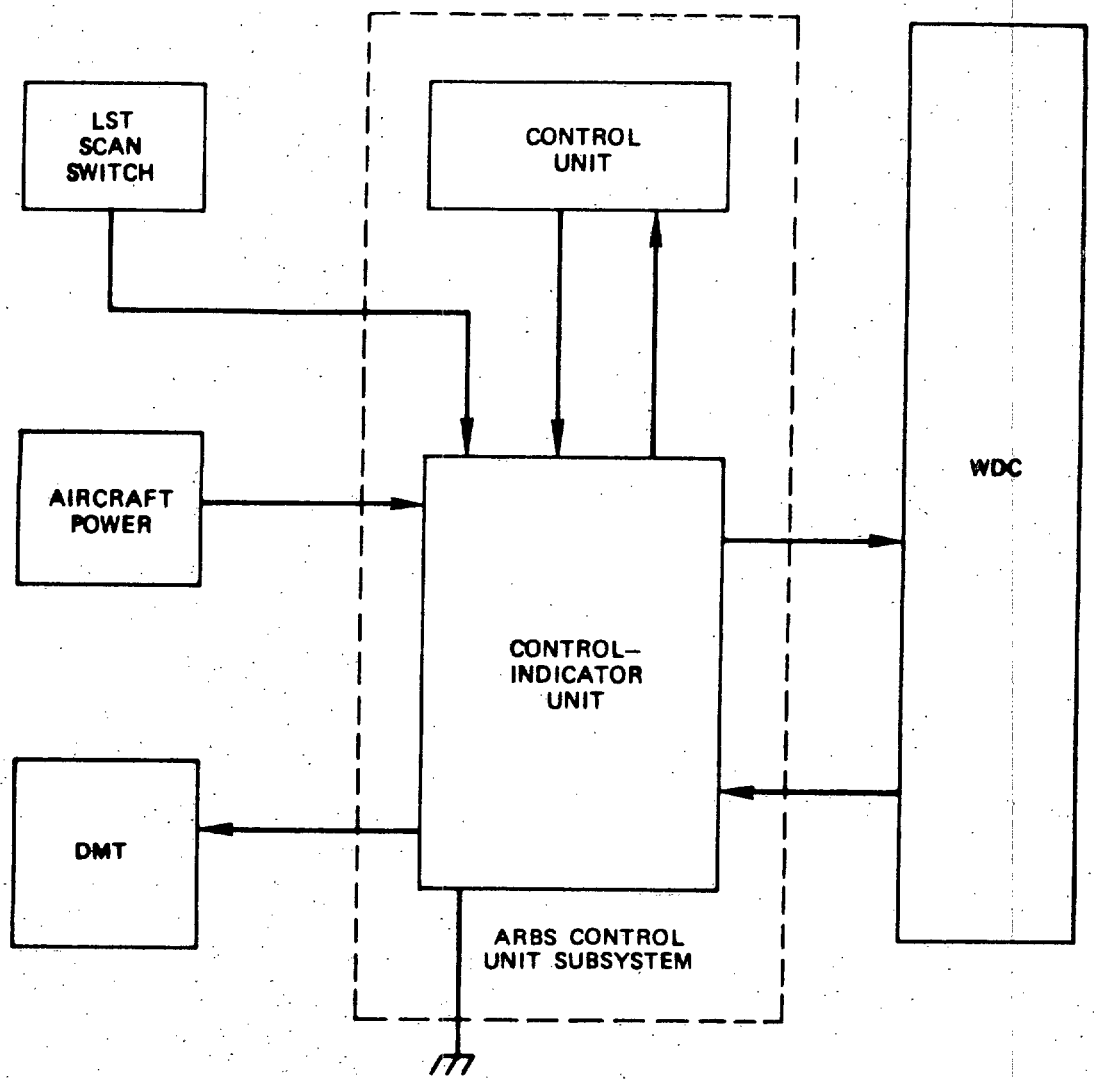


Figure 3. ARBS Control Unit Subsystem Functional Block Diagram.

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3.3.13.1.2 AV-8A Aircraft - The subsystem shall interface with the AV-8 aircraft in accordance with the provisions of MIL-A-85055.

3.3.13.1.3 WDC Interface - The subsystem shall interface with the WDC, as shown in Figure 4, through a 50-kilohertz (kHz) I/O serial data channel and be capable of I/O of data at a rate of 2.5 thousand (k) words/second. The signal timing between the subsystem and WDC shall be in accordance with Figure 5 and the I/O signals as described in Tables II and III. The subsystem shall supply 28 Vdc, with characteristics in accordance with MIL-STD-704, to the WDC for power turn ON control. The 50-kHz signal definition shall be in accordance with MIL-A-85055.

3.3.13.3 DMT Interface - The subsystem shall supply 28 Vdc, with characteristics in accordance with MIL-STD-704, to the DMT for power ON control.

3.3.13.4 LST Scan Switch Interface - The ARBS Control Unit Subsystem-LST Scan Switch interface shall be as shown on Figure 6, in accordance with MIL-A-85055.

3.4 Performance - The subsystem shall provide data entry/readout functions, display panel mode/data selection, and self-test and caution indications as delineated herein when operating in conjunction with other elements of the angle rate bombing set (ARBS) under standard conditions of 3.3.9, service conditions of 3.3.10, and power conditions of 3.3.12.

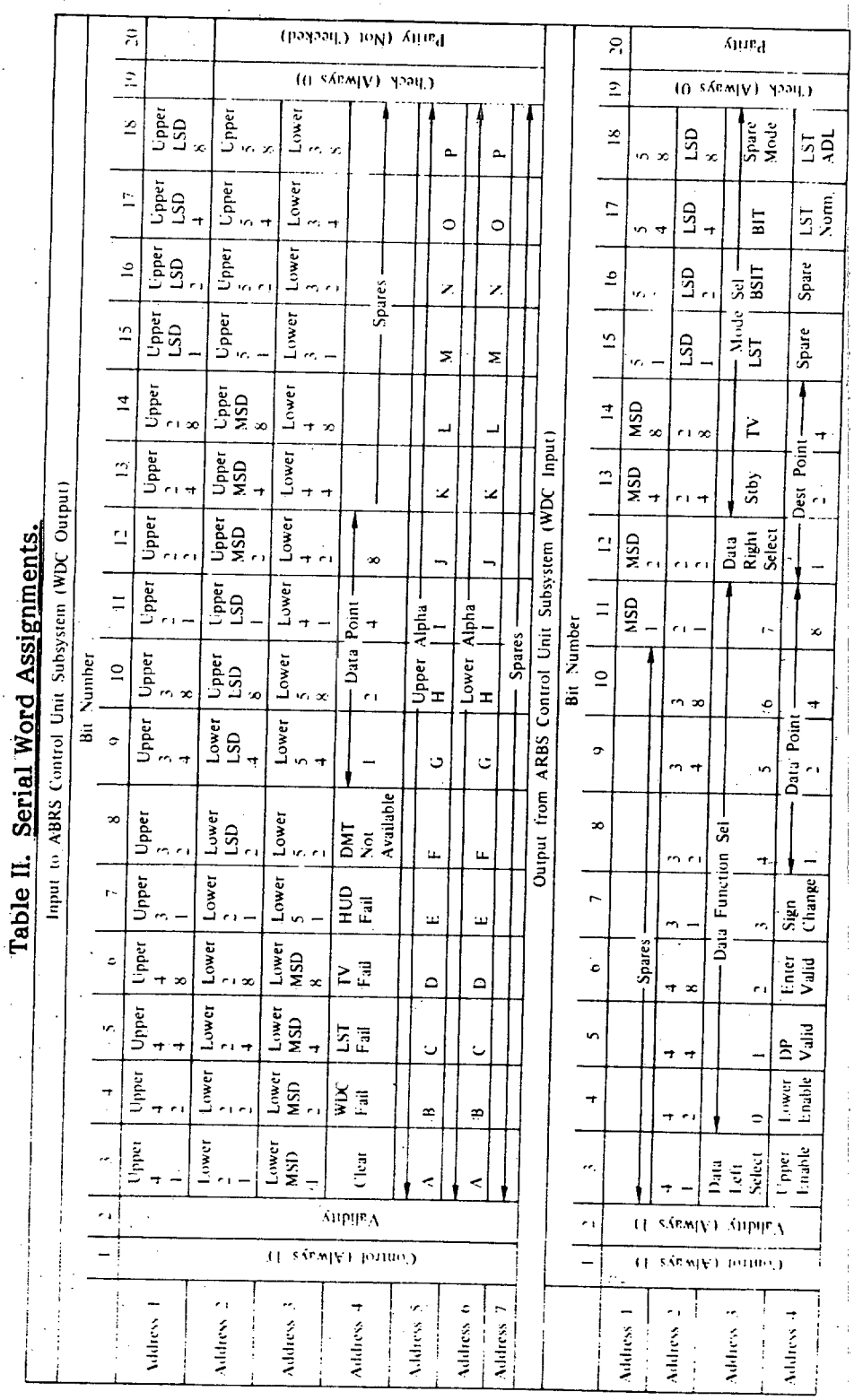
3.4.1. ARBS Control Unit - The Control, Bombing Set, Angle Rate C-10122/ASB-19(V) shall perform the functions specified by the following paragraphs.

3.4.1.1 System Control - The Control Unit (U) shall provide controls for Mode and Destination Point selection .

3.4.1.2 System Displays - The CU shall provide caution indications for the TV, LST, WDC, HUD, and DMT functions.

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Table II. Serial Word Assignments.



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Table III. WDC/Control Unit Subsystem I/O Interface.

Item No.	Signal Name	Characteristic
	WDC Input Signals From Control Unit Subsystem	
1.	DATA READY (OUT) and DATA READY (OUT) COMPLEMENT	Bit 1 will be synchronous with the rising edge of CLOCK pulse.
2.	DATA ADDRESS (OUT) and DATA ADDRESS (OUT) COMPLEMENT	Bit 1 will be synchronous with Bit 1 of DATA READY (OUT) and rising edge of CLOCK pulse and have the following addresses:

		ADDRESS CODE			
WORD	Bit	18	17	16	15
Address 1		0	0	0	1
Address 2		0	0	1	0
Address 3		0	0	1	1
Address 4		0	1	0	0

Bits 3 to 18 of the discrete words addressed 1 to 4 shall be as shown in Table II.

3.	WDC ON and WDC ON RETURN	28 Vdc turn ON control power to the WDC.
4.	DATA (OUT) and DATA (OUT) COMPLEMENT	Bit 1 will be synchronous with Bit 1 of DATA READY (OUT) and DATA ADDRESS (OUT) and rising edge of CLOCK pulse. Data will contain four 20-Bit serial data words with the addresses as specified in item 2.
5.	DATA READY (IN) and DATA READY (IN) COMPLEMENT	Bit 1 shall be synchronous with rising edge of CLOCK pulse. This bit, ANDed with Bits 1 of item 6 and item 7 of Table II, define the beginning of a data word.

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Table III. WDC/Control Unit Subsystem I/O Interface (Contd.)

Item No.	Signal name	Characteristic
		WDC Output Signals to the Control Unit Subsystem
6.	DATA ADDRESS (IN) and DATA ADDRESS (IN) COMPLEMENT	Bit 1 shall be synchronous with Bit 1 of DATA READY and the rising edge of the CLOCK pulse. The data address work shall contain the following addresses:

ADDRESS CODE

WORD	Bit	18	17	16	15
Address 1		0	0	0	1
Address 2		0	0	1	0
Address 3		0	0	1	1
Address 4		0	1	0	0
Address 5		0	1	0	1
Address 6		0	1	1	0
Address 7		0	1	1	1

Bits 3 to 18 of the discrete words addressed 1 to 7 shall be as shown in Table II.

7.	DATA (IN) and DATA (IN) COMPLEMENT	Bit 1 shall be synchronous with Bit 1 of DATA READY and DATA ADDRESS and rising edge of CLOCK pulse. DATA shall consist of seven 20-Bit serial words with addresses as specified in item 6.
8.	CLOCK and CLOCK COMPLEMENT	50 kHz timing clock for all digital signals.
9.	DATA INITIATE and DATA INITIATE COMPLEMENT	A low to high edge requests that all data words be transmitted by the subsystem to WDC.
10.	5 Vdc	Discrete voltage-power for subsystem digital circuits.

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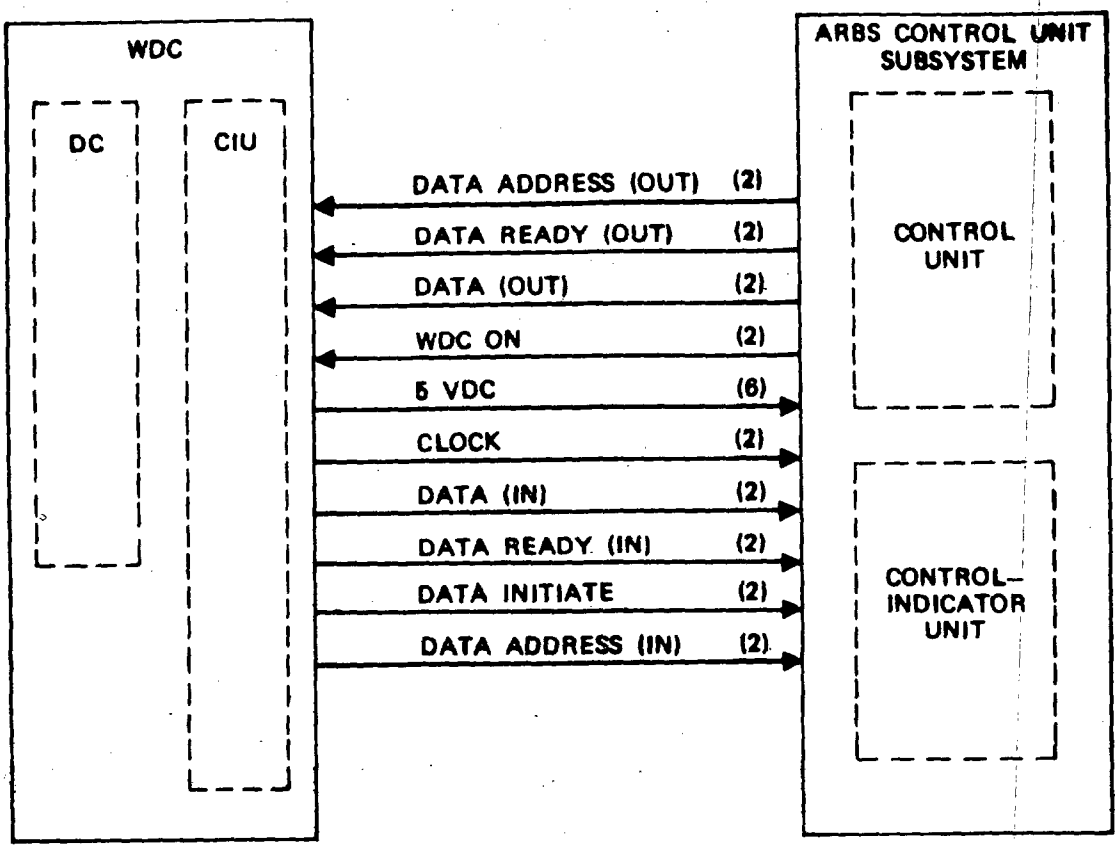


Figure 4. WDC/ARBS Control Unit Subsystem Interface.

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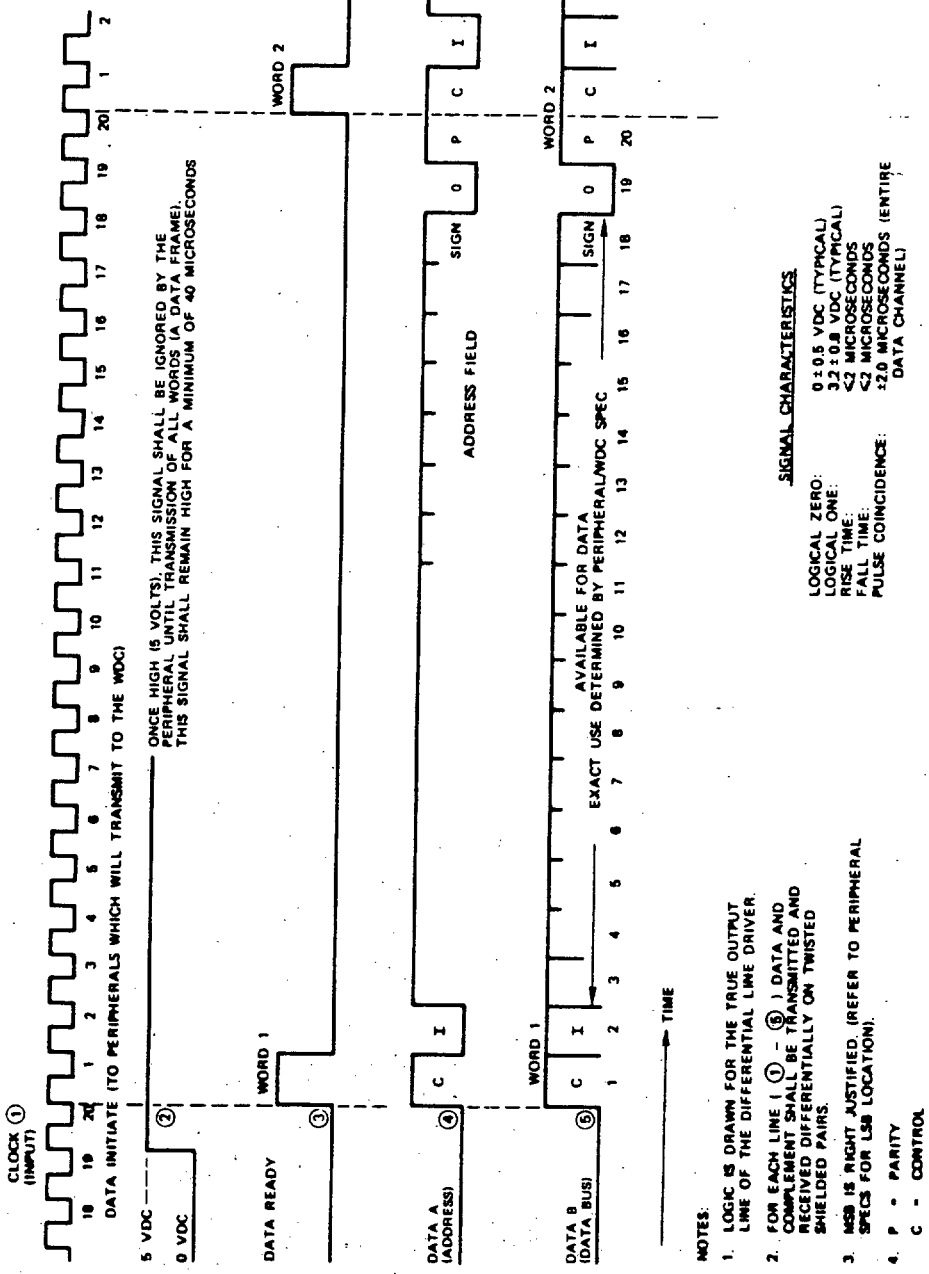


Figure 5. Signal Timing and Digital Format.

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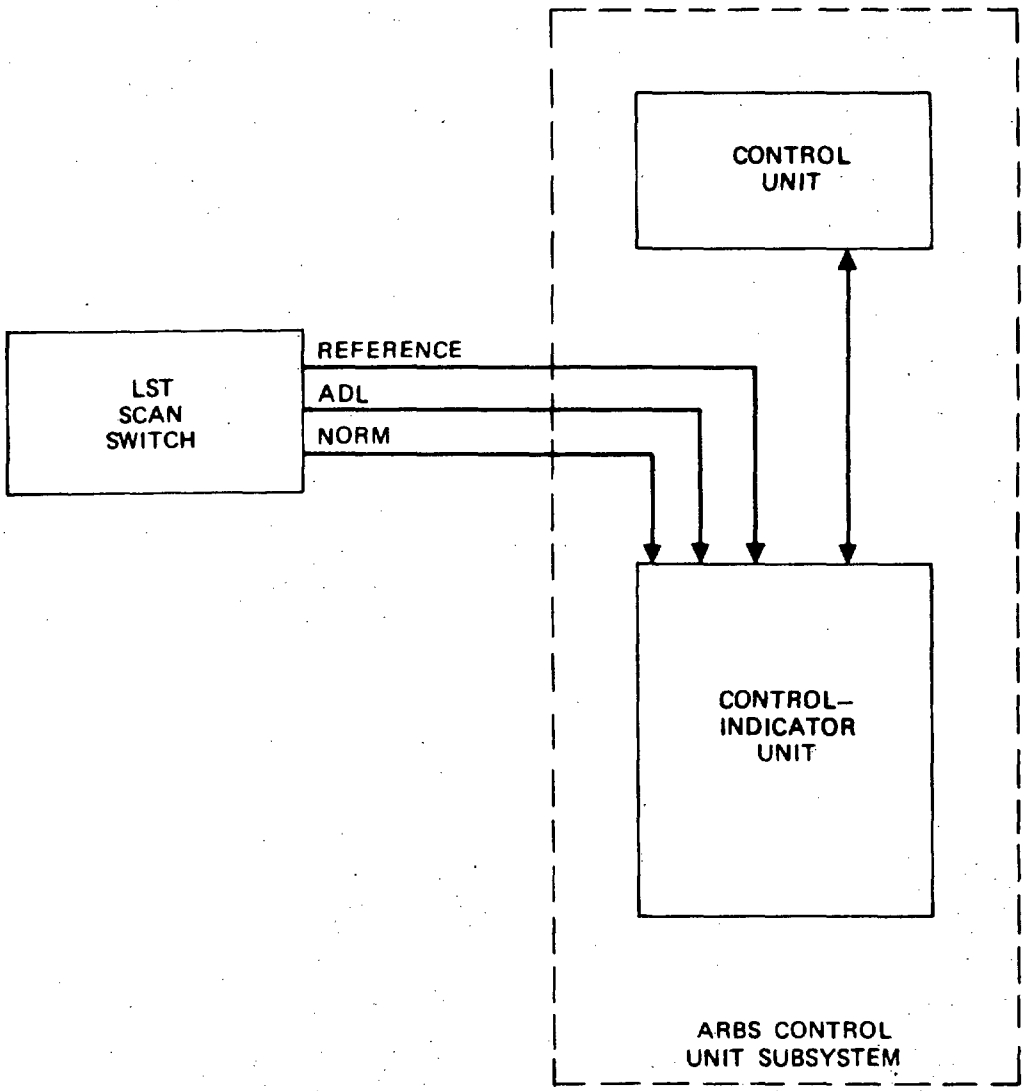


Figure 6. Subsystem-LST Scan Switch Interface.

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3.4.2 Data Entry Unit - The Control-Indicator, ID-2085/ASB-19(V) shall perform the functions as specified in the following paragraphs.

3.4.2.1 Data Entry and Readout - The Control-Indicator provides the capability for formatting and transmitting serial output data to the Weapons Delivery Computer (WDC) and receive/store/display serial input data from the computer as shown in Table II. Register readouts shall display data in associated pairs as shown in Table IV for the applicable Data Select switch positions and destination point selected.

3.4.2.2 Destination Points - The subsystem shall provide for entry and readout of target information on at least two targets and shall include provisions for entry and readout of data for at least five other destination points. One destination point shall be reserved as a MARK (MK) point and one as HOME BASE (HB).

3.4.2.3 Navigation Update - The Control-Indicator shall provide two selectable modes of navigation updating.

3.4.2.4 Keyboard - Data shall be entered through a 14-key keyboard with the capability of entering Clear, Sign Change, Enter, Display Selection, and Numerics for any of the functions listed in Table IV.

3.5 Detail Requirements - Detail requirements of the subsystem shall be as specified herein.

3.5.1 Item Definition - The ARBS Control Unit Subsystem, as part of the ARBS, shall provide the control and display functions required during operation of the Dual Mode Tracker (DMT) and WDC. The control panel general requirements shall be in accordance with MIL-C-81774 except as approved by the procuring activity. The subsystem shall provide the following functions:

(1) Data entry to and data display from the WDC based on data supplied from the cockpit by the pilot.

(2) Provide 28 volts direct current (Vdc) to WDC and DMT for power ON control.

(3) Selection of operating modes and the selection of targets and destination points.

(4) Indications of ARBS and other system failures and caution indications.

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Table IV. Data Entry/Readout Functions.

Display/entry functional requirement	Function switch legend	Display1	Implied scaling
Target elevation above sea level	Alt	YXXX X X X	Feet
Mean sea level pressure	MSLP	YXXX X.X X	Hundredths of inches of mercury
Loft mode identification point range to target	RNG	YXXX X X X	Feet
Loft mode identification point bearing to target	BRG	YXXX°X X.X	Degrees, tenths of minutes
Loft mode release angle	LOFT	YXXX°X X.X	Degrees, tenths of minutes
Loft mode pullup range to target	PUR	YXXX X X X	Feet
Chosen release range	CRR	YXXX X X X	Feet
(Spare)	- - -	YXXX X X X	Units (Spare)
Wind speed	WIND (Spd)	YXXX X X X	Knots
Wind direction	(Dir)	YXXX°X X.X	Degrees, tenths of minutes
Laser coding of the three digit code	CODE	YXXX X X X	Octal units
(Spare)	- - -	YXXX X X X	Units (Spare)
Distance to destination	RNG	YXXX X X X	Tenths of nautical miles
Time-to-go to destination	TTG	YXXX X:X X	Minutes, seconds
Present position latitude	POS (Lat)	YXXX°X X.X	Degrees, tenths of minutes
Present position longitude	(Long)	YXXX°X X.X	Degrees, tenths of minutes
Destination latitude	DEST (Lat)	YXXX°X X.X	Degrees, tenths of minutes
Destination longitude	(Long)	YXXX°X X.X	Degrees, tenths of minutes

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Table IV. Data Entry/Readout Functions (Contd.)

Display/entry functional requirement	Function switch legend	Display[1]	Implied scaling
Selection provision for other variables	DATA (Nbr)	YXXX X X X	Units
Data provision for other variables[2]	(Value)	YXXX X X X	Units (Scaling depends on parameter)
Weapon burst height	ΔB	YXXX X X X	Feet
Fuze delay time	FUZE	YXXX X.X X	Hundredths of seconds
Julian date		YXXX X:X X	
Greenwich mean time	GMT	YXXX X:X X	
Mark point latitude	MARK	YXXX°X X.X	Degrees, tenths of minutes
Mark point longitude		YXXX°X X.X	Degrees, tenths of minutes
HUD update mode - latitude	HUD UPDATE	YXXX°X X.X	Degrees, tenths of minutes
HUD update mode - longitude		YXXX°X X.X	Degrees, tenths of minutes
Flyover update mode - latitude	FO UPDATE	YXXX°X X.X	Degrees, tenths of minutes
Flyover update mode - longitude		YXXX°X X.X	Degrees, tenths of minutes
Code number of destination point values being displayed	DP	X	Decimal code

[1] Y = Blank or any combination of a 16-element alpha display.
X = Blank or any decimal formed on a 7-segment display.

[2] Maintenance biases, scale factors, constants, and other selected variables. The number of these data values selectable shall be limited only by the display.

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3.5.2 The Control Unit Subsystem shall conform to the requirements specified herein and shall consist of two units as shown in Figure 7.

3.5.2.1 Panel Configuration - The control and display panels shall be in accordance with Figure 7 and provide the following mode/data selection/display capability.

(1) ARBS Mode Select Switch (MODE).

(2) Self-test and caution indicators.

(3) Fly-to Destination Point Selection Switch (DEST PT).

(4) Numeric displays.

(5) Display brightness control (BRT).

(6) Data entry keyboard.

(7) Data select switches (DATA SEL).

3.5.2.1.1 Keyboard - A 14-key keyboard assembly shall be used for entering data. Numeric entry shall be provided for the destination point (DP) and data parameters provided on the data select switches. The keyboard format shall utilize a 2 x 5 decimal matrix and be as shown in Figure 7.

3.5.2.2 Readout Registers - Data to be entered or read out shall be represented on two identical numeric readout registers. The upper registers shall display the first of a two-parameter grouping (e.g., range and bearing). A single parameter shall be displayed in the upper register. Each register shall have an alphanumeric readout in the most-significant position followed by six numeric only readouts. There shall be three positions, after the third, fourth, and fifth numerics, that shall display either a decimal, degree sign, colon, or be blank as a function of switch position defined in Table III. As new data is entered, the display shall show the data entered. A single numeric only readout shall be provided for the selected data point. The readout registers shall have sufficient brightness and/or contrast for visibility under daylight ambient lighting conditions and shall have a uniformity of brightness better than 2.5:1.

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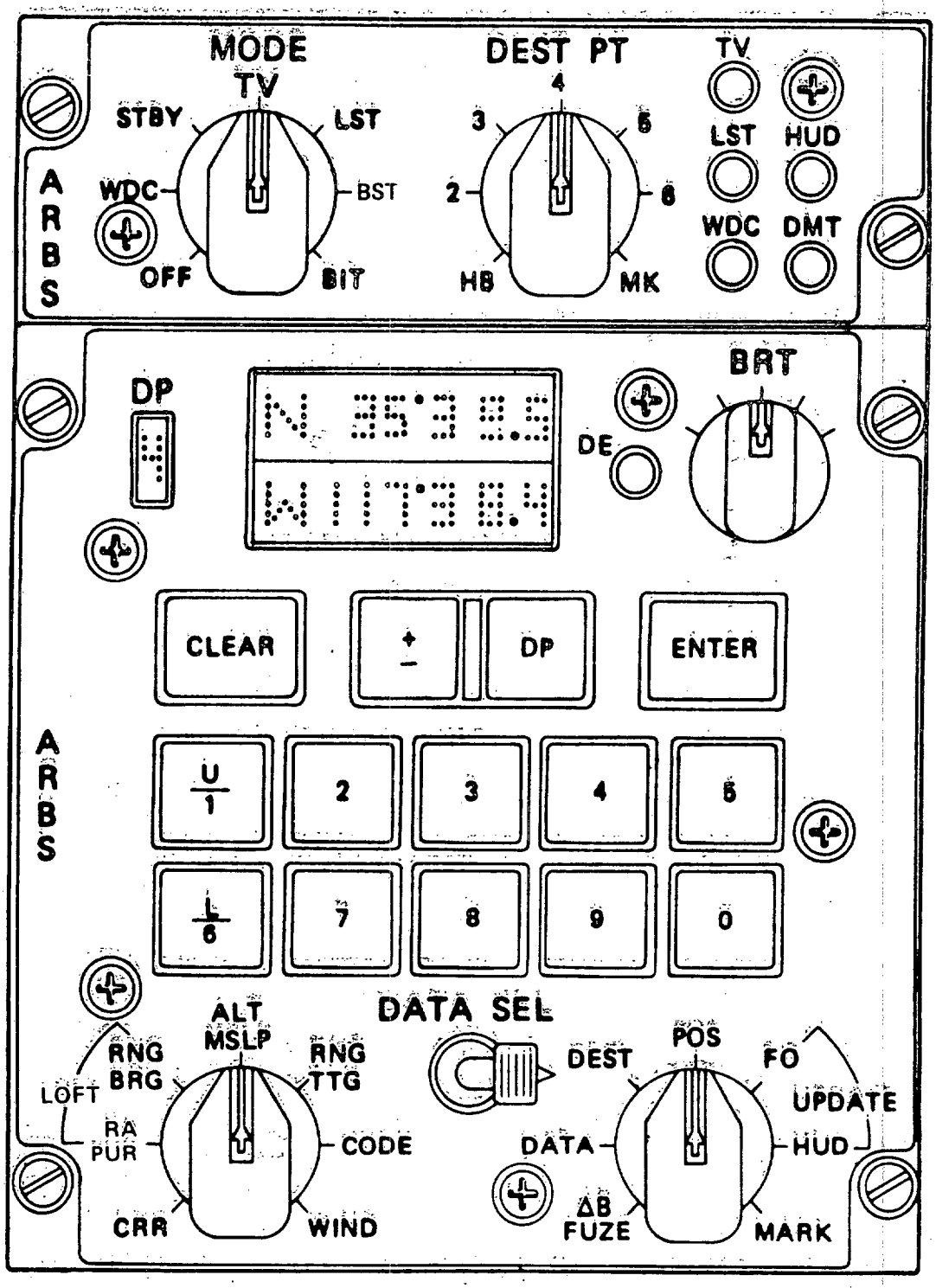


Figure 7. Subsystem Control and Display Panels.

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3.5.2.2.1 Alphanumeric Segment Identification Code - The alphanumeric segment identification code shall be as shown in Figure 8 and shall have the bit numbers and addresses as specified in Table II.

3.5.2.2.2 Numeric Code - The numeric only readouts shall have binary-decimal decoder drivers. The binary coding and addresses shall be as specified in Table II.

3.5.2.2.3 Display Format - The alphanumeric readouts shall be composed of 16 segments, and the numeric only readouts shall be composed of 7 segments with each segment individually illuminated by an incandescent lamp.

3.5.2.2.4 Display Selection - The first repression and release of either display selection button; digit one (1) for upper or digit six (6) for lower, shall clear the display and establish the input mode. Numerics entered will appear in the right-most (least significant digit) position and be shifted left on entry of next numeric. If CLEAR is selected or any rotary switch position changed, including the right/left toggle, the display will blank and revert to display input words. The clear bit in the input word shall override operator selected mode. The sign change shall be implemented as a full duplex mode. All switch actuations shall result in appropriate bits being set in the output words.

3.5.2.3 ARBS Mode Select Switch - The ARBS Mode Select Switch shall be an eight-position make-before-break rotary switch providing manual selection of the following modes:

	Mode	Function
(1)	OFF	28 Vdc ON-POWER removed from WDC and DMT.
(2)	WDC	28 Vdc applied to WDC only for turn-on control.
(3)	STANDBY	28 Vdc applied to WDC (STBY) and 28 Vdc applied to DMT for turnon control and request to WDC for Standby Mode.
(4)	TV	28 Vdc applied to WDC and DMT and request to WDC for Television Tracker Mode.
(5)	LST	28 Vdc applied to WDC and DMT, and request to WDC for Laser Spot Tracker Mode.

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(6)	BST	28 Vdc applied to WDC and DMT and request to WDC for the Boresight Mode which enables in-flight adjustment of the boresight between the HUD aiming reticle and DMT line-of-sight.
(7)	BIT	28 Vdc applied to WDC and DMT and request to WDC for the Spare Mode.
(8)	SPARE	28 Vdc applied to WDC and DMT and request to WDC for the Spare Mode.

3.5.2.4 Self-Test and Caution Indications - Six yellow indicator lights shall be provided for test, failure, and caution indications of the WDC, TV, LST, DMT, HUD, and DE. The caution indicators shall be illuminated whenever the appropriate bits are set in the input words shown in Table II. The DE yellow light shall indicate whenever the unit register readouts are under operator control and not providing updated computer data. The indicator lights shall illuminate a steady yellow color when activated and have sufficient brightness for visibility under daylight ambient lighting conditions.

3.5.2.5 Destination Selection - The subsystem shall provide a rotary switch which has selectable positions for five targets or navigation points, HB and MK. These selections will be used to request data from the WDC. These same points shall be selectable for data display and readout via an independent control.

3.5.2.6 Display Brightness Trimming Control - The display brightness will be controlled by the aircraft lighting system. The subsystem shall provide a trimming control designated BRT with five discrete selections of display brightness. During day operation, maximum display brightness shall be obtained by a maximum clockwise setting. Normal brightness selection shall be straight up as indicated by the arrow on the control knob. The maximum counterclockwise selection shall provide a maximum trimming of the display brightness. The maximum and minimum settings shall provide a brightness variation from normal of at least 2:1 during night operation.

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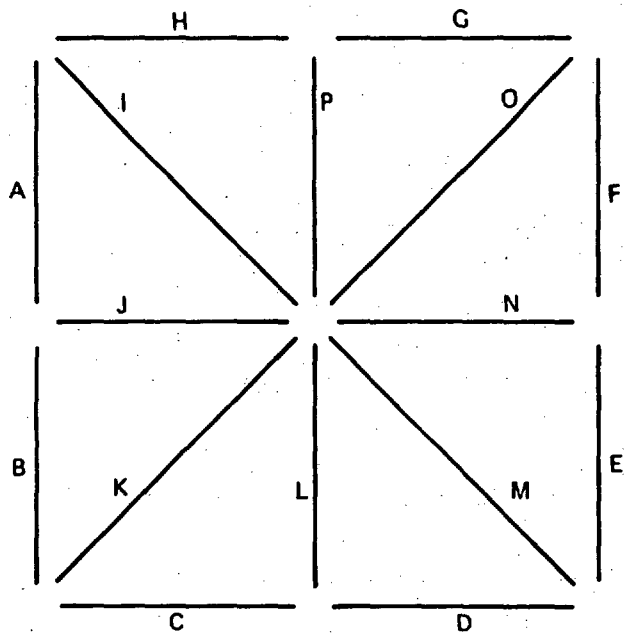


Figure 8. Alphanumeric Segment Identification Code.

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3.5.2.7 Data Selection Switches - The subsystem shall provide a two-position toggle switch, designated DATA SEL, and two rotary switches containing a total of 16 data selection positions. The switches shall have the following functions:

(1) Two-position toggle switch, sets bits indicated in Table II, and in:

(a) Left position - Selects rotary switch on left side of panel for data entry/readout.

(b) Right position - Selects rotary switch on right side of panel for data entry/readout.

(2) Left side rotary switch.

(a) CRR position selected - Provides display/entry in the upper readout register of chosen release range in feet.

(b) RA / PUR position selected - (LOFT MODE) Provides display/entry in the upper readout register of loft mode release angle in degrees and tenths of minutes, and display/entry in the lower readout register of loft mode pull-up range in feet.

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(c) RNG / BRG position selected (LOFT MODE) - Provides display/entry in the upper readout register of identification point to target range in feet, and display/entry in the lower readout register of the identification point to target bearing in degrees and tenths of minutes.

(d) ALT / MSLP position selected - Provides display/entry in the upper readout register of target elevation above sea level in feet, and display/entry in the lower readout register of MSL pressure at the designated target in hundredths of inches of mercury.

(e) RNG / TTG position selected - Position will provide display of the range, upper readout register, in tenths of nautical miles and the time to-go, lower readout register, in minutes and seconds to the selected destination point when future navigation growth of the WDC has been implemented.

(f) CODE position selected - Provides display/entry in the upper readout register of the three-digit Laser Spot Tracker Code being used.

(g) WIND position selected - Provides display/entry in the upper readout register of the wind speed in knots, and display/entry in the lower readout register of wind direction in degrees and tenths of minutes.

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(h)	Spare position-	Provided for growth.

(3) Right side rotary switch.

(a)	$\frac{\Delta B}{FUZE}$ position selected -	Provides display/entry in the upper readout register of the weapon burst height in feet, and display/entry in the lower readout register of the fuze delay time in hundredths of seconds.
(b)	DATA position selected -	Provides display/entry in the upper readout register of a decimal code number which is used for storing and calling up special parameters not provided as selectable discrete switch positions (e.g., magnetic variation, ground speed, ground track, laser search antenna angles, maintenance biases, scale factors, and constants). The lower readout register shall be used for display/entry of these special parameters.
(c)	DEST position selected -	Position will provide display/entry of destination point latitude and longitude in degrees and tenths of minutes when future navigation growth of the WDC has been implemented.

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(d)	POS position selected -	Position will provide display of own aircraft present latitude and longitude in degrees and tenths of minutes when future navigation growth of the WDC has been implemented.
(e)	FO UPDATE position selected -	Position will provide display of latitude and longitude of destination point, and enables the flight stick destination control for use in updating the navigation system in the flyover mode (in degrees and tenths of minutes) when future navigation growth of the WDC has been implemented.
(f)	HUD UPDATE position selected -	Position will provide display for latitude and longitude of destination point, and enables the flight stick designation control for use in updating the navigation system in the HUD mode (in degrees and tenths of minutes) when future navigation growth of the WDC has been implemented.
(g)	MK position selected -	Position will provide display/entry of latitude and longitude of MK Point and enables the designation control for entering a MK Point (in degrees and tenths of minutes) when future navigation growth of the WDC has been implemented.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the contractor is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests listed in Tables V through VIII. To determine compliance with all applicable requirements (see Figure 9)

(1) Preproduction (First Article) Tests (see 4.2)

(2) Acceptance Tests (see 4.3)

(3) System Performance Demonstration Tests (see 4.4)

(4) Special Tests (see 4.5)

4.1.2 Special Working Environment - The contractor shall provide adequate facilities for the fabrication, assembly and testing of the items covered be this specification, in accordance with MIL-A-85055.

4.1.3 Government Verifications - Government verification shall be in accordance with MIL-A-85055.

4.1.4 Test Conditions - Unless otherwise specified, all tests for acceptance shall be performed at the standard conditions specified by MIL-A-85055.

4.1.5 Spare Equipment Provisioning (WRA/SRA) Acceptance - Unless otherwise specified by the contract, it is intended that the items covered by this specification shall be accepted at the subsystem (CUS)/ system (ARBS) level of assembly, in accordance with MIL-A-85055, to ensure that the equipment interchangeability and maintainability requirements of MIL-A-85055 are met.

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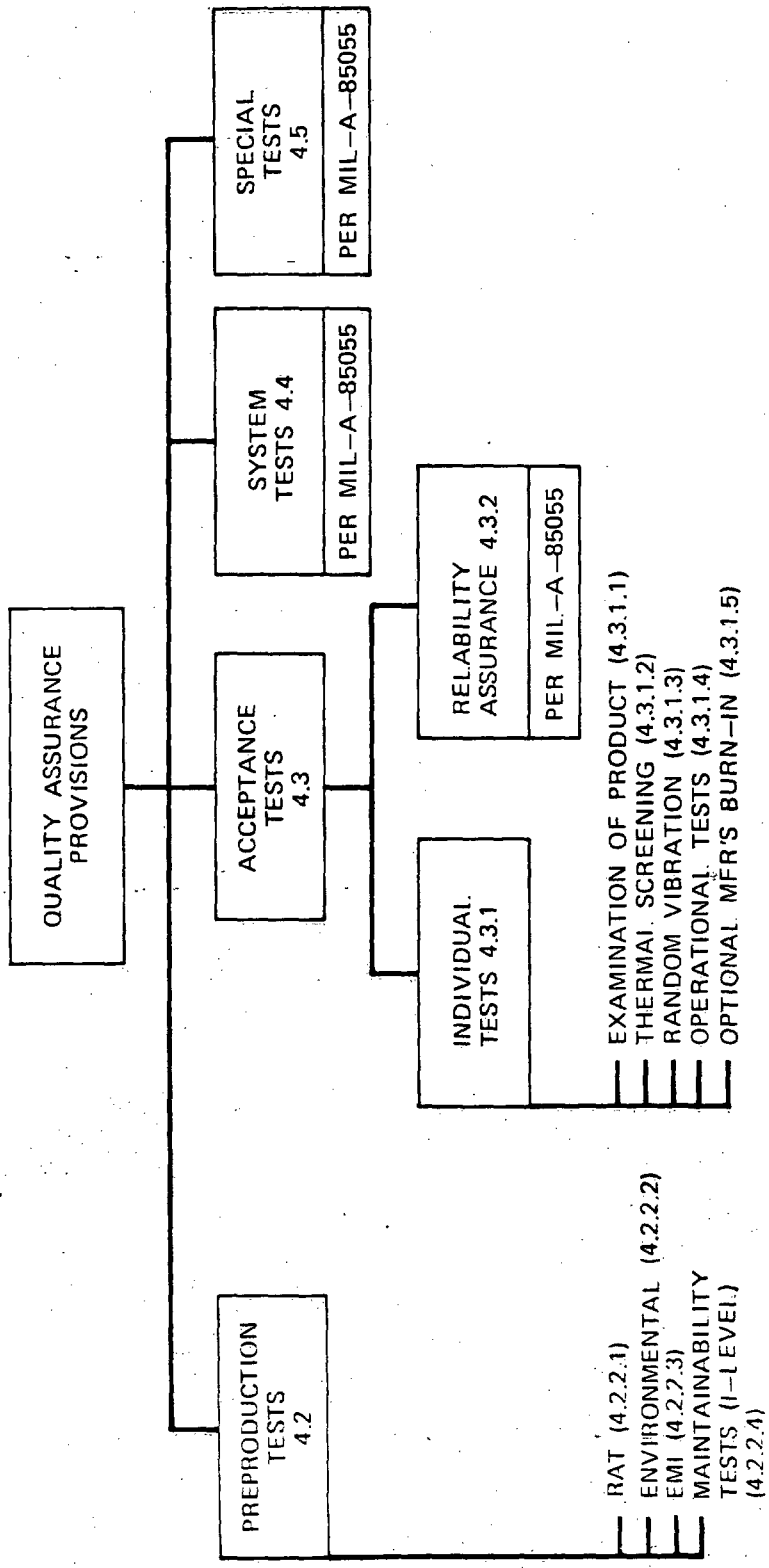


Figure 9. Classification of Tests.

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Table V. Quality Assurance Test Requirements.

REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	TITLE	PREPRODUCTION TESTS	ACCEPTANCE TESTS
3.3.10	4.2.2.2, 4.6.1	Environmental Tests	X	
3.3.2, MIL-D-85055	4.2.2.1	Reliability Assurance Tests	X	
3.3.6	4.2.2.3	Electro-magnetic Interference Tests	X	
3.3.5, 3.3.7 MIL-A-85055	4.2.2.4	Maintainability Demo (I-Level)	X	
3.2, 3.3, 3.3.4, 3.3.8	4.3.1.1	Examination of Product	X	X
MIL-A-85055	4.3.1.2	Thermal Screening	X	X
MIL-A-85055	4.3.1.3	Random Vibration	X	X
3.4, 3.5	4.3.1.4, 4.6.2	Operational Tests	X	X
OPTIONAL	4.3.1.5	Optional Mfr's Burn-in	X	X
3.3.2	4.3.2	Reliability Assurance Tests		X
MIL-A-85055	4.4	System Tests	X	X
MIL-A-85055	4.5	Special Tests (AR = As Req'd	AR	AR

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Table VI. Environmental Test Requirements.

REQUIREMENT PARAGRAPH	METHOD PARAGRAPH	TITLE
3.3.10.1, 3.3.10.1	4.6.1.1	Low Temperature
3.3.10.1, 3.3.10.2	4.6.1.2	High Temperature
3.3.10.1	4.6.1.3	Temperature Shock
3.3.10.2	4.6.1.4	Low Pressure/Altitude Test
3.3.10.3	4.6.1.5	Humidity
3.3.10.4	4.6.1.6	Vibration
3.3.10.5	4.6.1.7	Shock
3.3.10.6	4.6.1.8	Gunfire Vibration
3.3.10.7	4.6.1.9	Acoustic Noise
3.3.10.8	4.6.1.10	Salt Spray
3.3.10.9	4.6.1.11	Sand and Dust
3.3.10.10	4.6.1.12	Explosive Atmosphere
3.3.10.11	4.6.1.13	Fungus
3.3.11	4.6.1.14	Warm-up Time
3.3.12	4.6.1.15	Operating Power

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Table VII. Examination of Product Requirements.

REQUIREMENT PARAGRAPH	TITLE	PREPRODUCTION ACCEPTANCE	PRODUCTION ACCEPTANCE
MIL-A-85055			
3.2.4	Mainainability Provisions	X	
3.3	Design/Construction (MIL-E-5400)	X	
3.3.1	Parts and Materials (MIL-E-5400)	X	
3.3.1.2	Connectors	X	X
3.3.1.3	Adjuctment Controls	X	
3.3.1.4	Printed Wiring Boards	X	X
3.3.1.5	Printed Wiring Assemblies	X	X
3.3.1.6	Elastomeric Materials	X	
3.3.1.7	Soldering	X	X
3.3.1.8	Wiring/Wire Coding	X	
3.3.1.9	Surface Finishes	X	X
3.3.1.10	Logic Circuits	X	
3.3.1.11	Screws and Fasteners	X	X
3.3.1.12	Vibration Isolation	X	
3.3.1.13	Accessibility	X	
3.3.2.5	Bonding and Grounding	X	X
3.3.3	Nomenclature/Nameplates/ID	X	
3.3.3.1	Serialization	X	
3.3.4	Workmanship	X	X
3.3.4.1	Cleanup	X	X
3.3.5	Interchangeability	X	

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Table VII. Examination of Product Requirements (Continued).

REQUIREMENT PARAGRAPH	TITLE	PREPRODUCTION ACCEPTANCE	PRODUCTION ACCEPTANCE
MIL-A-85055 (CONT)			
3.3.8	Modules	X	
3.3.10	Cooling	X	
3.3.12	Boresight Provisions	X	
MIL-C-85058			
3.3.1.1	Total Weight	X	
3.3.1.2	Dimensions	X	X
3.3.2.4	Time Totalizing Meter	X	X
3.3.3	Connectors	X	
3.3,7	Nomenclature		X
3.3.8	Serialization		X

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Table VIII. Quality Assurance Performance (Operational) Tests.

REQUIREMENTS PARAGRAPH	METHOD PARAGRAPH	TITLE	PREPRODUCTION TESTS	ACCEPTANCE TESTS
3.4.1.1	4.6.2.1	Data Entry and Readout	X	
3.4.1.2	4.6.2.1	Data Points	X	
3.4.1.3	4.6.2.1	Navigation Update	X	
3.4.1.4	4.6.2.2	System Control	X	
3.4.1.5	4.6.2.3	System Displays	X	
3.4.1.6	4.6.2.3	Keyboard	X	
3.5.2.1	4.6.1.1	Panel Configuration (Examination)	X	
3.5.1.1	4.6.2.1,4.6.2.4	External Interfaces	X	
3.3.12.4, 3.3.12.5	4.6.2.5	Overload Protection (Certification)	X	
3.5.1.1.1	4.6.2, 4.6.2.4	External Interface		X
3.4.1, 3.5.2	4.6.2.1	Keyboard Entry		X
3.5.2.3, 3.5.2.4	4.6.2.2	ARBS Mode Select Switch		X
3.5.2.4	4.6.2.3(1)	Self-test and Caution Indicators		X
3.5.2.6	4.6.2.3(2)	Display Brightness Control		X

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4.1.6 Manufacturing Reliability Screening - Each item covered by this specification shall be subjected to manufacturing reliability screening for the purpose of screening defective workmanship and parts, in accordance with MIL-A-85055.

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipment to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commerical testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review, in accordance with MIL-A-85055.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include:

(1) Individual Tests (see 4.3.1).

(2) Reliability Assurance Tests (see 4.2.2.1).

(3) Environmental Tests (see 4.2.2.2, 4.6.1).

(4) Electromagnetic Interference Tests (see 4.2.2.3).

(5) Maintainability Demonstration I-Level (see 4.2.2.4).

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4.2.2.1 Preproduction Reliability Assurance Tests (RAT) - Preproduction RAT shall be conducted in accordance with MIL-A-85055.

4.2.2.2 Environmental Tests - Environmental tests shall be performed in accordance with the procedure requirements of 4.6.1, to ensure that the requirements of 3.3.9, 3.3.10 3.3.11 and 3.3.12 are met. The environmental test requirements are listed in Table VI. The test sequence shall be determined by the contractor, subject to approval by the procuring activity.

4.2.2.3 Electro-magnetic Interference (EMI) Tests - EMI tests shall be performed in accordance with MIL-A-85055, as specified in 4.6.3.

4.2.2.4 Maintainability Demonstration Tests (I-Level - The first level (I-level) maintainability requirements specified by 3.3.4, 3.3.6 and MIL-A-85055 shall be verified by a maintainability test. The test requirements and procedures shall be prepared by the contractor and submitted to the procuring activity for approval. The test procedures shall include procedures and requirements for adjustments and alignments required at all levels of assembly.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production subsystems shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. Disposition of the preproduction sample shall be as specified in the contract or purchase order (see 6.2).

4.2.4 Production - Subsystems supplied under the contract shall, in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each CUS shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection. For subsystems already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

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4.3 Acceptance Tests - The contractor shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such records shall be signed by an authorized representative of the contractor or laboratory, as applicable and shall be available upon request by the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests (see 4.3.1).

(2) Reliability Assurance Tests (see 4.3.2).

4.3.1 Individual Tests - Each CUS submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each CUS accepted shall have passed the following tests:

(1) Examination of Product (see 4.3.1.1 and Table VII).

(2) Thermal Screening (see 4.3.1.2).

(3) Random Vibration (see 4.3.1.3).

(4) Operational Tests (see 4.3.1.4 and Table VIII).

(5) Optional Manufacturer's Burn-in (see 4.3.1.5).

4.3.1.1 Examination of Product - Each CUS shall be examined carefully to determine that the material and workmanship requirements have been met. The tests shall consist of the physical and mechanical examinations listed in Table VII.

4.3.1.2 Thermal Screening - Each item covered by this specification shall be subjected to thermal screening tests as specified by MIL-A-85055.

4.3.1.3 Random Vibration Tests - Each item covered by this specification shall be subjected to random vibration tests in accordance with MIL-A-85055 for the purpose of screening workmanship defects. The tests shall be performed in accordance with test procedures prepared by the contractor and approved by the procuring activity.

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4.3.1.4 Operational Tests - They shall be subjected to the operational tests specified in Table VIII in accordance with test procedures prepared and submitted by the contractor and approved by the procuring activity. The test procedure requirements are outlined in 4.6.2. During the operational tests, each unit under test shall be operated long enough to permit temperature stabilization and to provide sufficient time to check applicable characteristics and to record adequate data to assure satisfactory operation. Acceptance operational tests shall be performed prior to and after the completion of the Reliability Assurance (All Equipment-Burn-in) tests.

4.3.1.5 Optional Manufacturer's Burn-in - Optional burn-in tests may be performed by the contractor in preparation for the reliability assurance tests.

4.3.2 Reliability Assurance Tests (RAT) - Each CUS submitted for acceptance shall be subjected to the hours of burn-in, and the test conditions specified by MIL-A-85055, with the last 24 consecutive hours of test being failure free for acceptance, in accordance with MIL-A-85055. RAT shall be conducted at the system (ARBS) level of assembly. The accept/reject criteria of MIL-STD-781B Test Plan XXIX shall be used for acceptance.

4.4 System Performance Demonstration Tests - Each CUS submitted for acceptance shall be functionally tested, in accordance with MIL-A-85055, when integrated with the associated ARBS equipment. The functional tests shall consist of demonstration of control and mode switching, data entry and readout capabilities and observations of lamp and numeric display operations.

4.5 Special Tests - Special tests shall be conducted as required by the procuring activity, in accordance with MIL-A-85055.

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4.6 Test Procedures - The procedures used for conducting all tests specified herein shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.6.1 Environmental Tests - Environmental tests shall be in accordance with the environmental test plan and the procedures of MIL-STD-810, except as modified herein.

4.6.1.1 Low Temperature Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 502. Low temperature limits shall be as specified in 3.3.10.1. During the operating portion of the low temperature cycle, the subsystem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4.

4.6.1.2 High Temperature Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 501, Procedure I. The nonoperating high temperature (Step 2) and operating high temperatures (Step 4) shall be in accordance with 3.3.10.1 During the operating portion of the high temperature cycle, the subsystem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4.

4.6.1.3 Temperature Shock Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 503, Procedure I, except the maximum temperature shall be +85°C. After completion of temperature cycling, the subsystem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4.

4.6.1.4 Altitude Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 500, Procedure II. The low pressure condition (Step 3) shall be equivalent to 45,000 feet. Minimum temperatures shall not be less than those specified in 3.3.10.1. During the operating portions of the test, the subsystem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4.

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4.6.1.5 Humidity Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 507, Procedure I as modified by the condition of Table I for a minimum of 240 hours. After removal from the test chamber and return to room ambient conditions, the subsystem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4.

4.6.1.6 Vibration Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 514.1, Procedure VII, Part 2 in each of three orthogonal axes. Vibration levels shall be as specified in Table I. During each phase of vibration testing, no data output shall occur. Upon completion of testing, the subsystem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4.

4.6.1.7 Shock Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 516.1, Procedure I. Shock levels shall be in accordance with the requirements of 3.3.10.4. Upon completion of shock testing, the subsystem shall be subjected to the operatio al tests of 4.6.2 through 4.6.2.4.

4.6.1.8 Gunfire - The subsytem shall be tested with simulated gunfire vibration in both the operating and nonoperating conditions in accordance with the requirements of MIL-STD-810, Method 519.1, Procedure I. During the simulated gunfire vibration, no data output shall occur. Upon completion of testing, the operational tests of 4.6.2 through 4.6.2.4 shall be performed.

4.6.1.9 Acoustic Noise Test - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 515.1, Procedure I, Category B to the limits specified in 3.3.10.10. During and after acoustic noise testing, the subsystem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4.

4.6.1.10 Salt Atmosphere - The subsystem shall be tested in accordance with the requirements of MIL-STD-810, Method 509, Procedure I. Upon completion of testing, the subsytem shall be subjected to the operational tests of 4.6.2 through 4.6.2.4

4.6.1.11 Sand and Dust - The subsystem shall be tested to the requirements of MIL-STD-810, Method 510, Procedure I. Upon completion of testing, the subsytem shall be removed from the test chamber and subjected to the operational tests of 4.6.2 through 4.6.2.4.

4.6.1.12 Explosive Atomsphere - The subsystem shall be tested in accordance with the test requirements of MIL-STD-810, Method 511, Procedure I. The operational performance test shall be performed during the exposure to the explosive environment. In lieu of testing, a certification that the subsystem meets the explosion proof requirement of 3.3.10.9 may be submitted.

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4.6.1.13 Fungus - A certification that the requirements for fungus inert materials and components have been met may be submitted in lieu of performing a fungus test.

4.6.1.14 Warm Up Time - The preproduction CUS shall be tested in accordance with an approved test plan, to verify that the requirements of 3.3.11 are met.

4.6.1.15 Operating Power - The preproduction CUS shall be tested in accordance with an approved test plan to verify that the requirements of 3.3.12 are met.

4.6.2 Operational Tests - Each subsystem shall be operated over a time period sufficiently long enough to permit equipment temperatures to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory subsystem operation. All tests shall be performed with nominal input electrical power applied except that the preproduction tests shall be performed with input electrical power at nominal and at the extremes specified in MIL-A-85055 and MIL-STD-704. The power shall be applied in accordance with 3.3.12 and with a reference 50-kHz clock signal applied. The operational requirements of the individual tests specified in Table V and shall be demonstrated including characteristics of the 50-kHz signal reception and transmission.

4.6.2.1 Data Entry and Readout Functions -

(1) Keyboard entry: Each of the keys on the 14-key keyboard assembly shall be activated in a logical sequence while observing the serial data outputs and the Subsystem Display Panel displays. The serial data outputs and the observed displays specified by 3.4.1.1 shall verify compliance with the applicable portions of 3.4.1 and 3.5.2 for each activation sequence.

(2) Destination point selection: The subsystem serial data outputs shall be observed for each position of the DEST PT. The serial data outputs shall conform to the requirements in 3.4.1 for each position to verify compliance with the applicable portions of 3.4.1.2 and 3.5.2.5.

(3) Data selection: The subsystem serial data outputs shall be observed for each position of the DATA SEL with the appropriate position of the Data Right and Left Select Switches. The serial data outputs shall conform to the requirements of 3.4.1.3 for each switch position to verify conformance to the applicable portions of 3.4.1, 3.4.1.1, 3.4.1.5, and 3.5.2.7.

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(4) Data entry: Serial inputs shall be applied to the subsystem in a logical sequence to exercise all input serial bit word assignments established in 3.4.1.1. The alphanumeric displays shall be observed for each input to verify the requirements of the applicable portions of 3.4.1, and 3.5.2.

4.6.2.2 Mode Selection -

(1) The voltage between WDC ON and DMT ON and 28-volt return shall be measured for all positions of the MODE switch. On all positions except OFF, the voltage measured on the WDC ON signal shall be 28 Vdc. On all positions except OFF and WDC, the voltage measured on the DMT ON signal shall be 28 Vdc.

(2) The MODE switch shall be sequentially switched to STBY, TV, LST, BSIT, and BIT positions while the format of the serial data output is observed. The serial data output format shall be in accordance with the requirements of 3.5.2.3(3) through (7).

4.6.2.3 Lamp Displays -

(1) With the MODE switch in the BIT position, serial data inputs shall be sequentially applied to the control subsystem to simulate the WDC outputs for WDC FAIL, LST FAIL, TV FAIL, HUD FAIL, and DMT NOT AVAIL in accordance with the requirements established by 3.4.1.5. The proper operation of each lamp shall be observed to be in conformance with 3.5.2.4 for the appropriate input serial data format (i.e., WDC Lamp On for WDC FAIL format, etc.).

(2) The average brightness of the alphanumeric displays shall be measured with both night and day voltages applied at the maximum, minimum, and nominal positions of the control subsystem brightness control (BRT) to assure compliance with 3.5.2.2, 3.5.2.6, and MIL-STD-1472.

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4.6.2.4 LST SCAN - With the mode select switch in the LST position, a LST Norm Scan signal and a LST ADL Scan signal shall be sequentially applied to the control subsystem while observing the serial data output format. The serial data output format shall be in accordance with the requirements established by 3.4.1.1 for requesting LST normal scan and LST armament datum line scan modes, respectively.

4.6.2.5 Circuit Protection - The overload protection requirements for parts and circuits of the subsystem shall be verified by a review of the circuit design analysis limited to areas which might create overload conditions. Requirements of 3.3.12.4 and 3.3.12.5 are satisfied when it is shown that suitable overload protection has been provided.

4.6.2.6 Internal Protection - Internal protection requirements of 3.3.12.6 shall be satisfied during the circuit design analysis.

4.6.3 Electromagnetic Interference Test - The subsystem shall be tested for EMI in accordance with MIL-A-85055.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part, or complete subsystem shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Any subsystem which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - The control unit subsystem shall be prepared for delivery in accordance with MIL-A-85055.

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6. NOTES

6.1 Intended Use - The subsystem is intended for control and display functions for operations of the DMT and WDC as part of the ARBS. The subsystem is intended for installation in the A-4M (ECP 1120 configuration) and MOD 800 for the AV-8A aircraft.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedences shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) System Specification - The system specification, MIL-A-85055, shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.6 Abbreviations - The abbreviations, acronyms, and symbols used herein are defined as follows:

ac	alternating current
ADL	armament datum line
ADR	address
ALT	target elevation
ARBS	angle rate bombing set
BIT	built-in-test
BRG	bearing
BST	boresight
BRT	brightness
C	control or Celsius
CLK	clock
CIU	computer interface unit
CRR	chosen release range
CUS	Control Unit Subsystem

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dB	decibels
dBm	decibels above (or below) 1 milliwatt
dc	direct current
DC	digital computer
DE	data entry
DEST	destination
DMT	dual mode tracker
DP	destination point
EMC	electromagnetic compatibility
EMI	electromagnetic interference
FO	flyover
g	gravitational constant
G	giga (10^9)
GEN	generator
GMT	greenwich mean time
GSE	ground support equipment
HB	home base
HUD	head-up display
I/O	input/output
k	kilo (10^3)
L, LWR	lower
LSB	least significant bit
LSD	least significant digit

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LST	laser spot tracker
LTG	lighting
M	mega (10^6)
MAX	maximum
MCT	maintenance correction time
MEA	maintenance engineering analysis
MK	mark
MSB	most significant bit
MSD	most significant digit
MSI	medium scale integration
MSLP	mean sea level pressure
MTBF	mean-time-between-failures
MTTR	mean-time-to-repair
NA	not applicable
NORM	normal
P	parity
PGSE	peculiar ground support equipment
POS	aircraft present position
PT	point
PUR	pullup range
RF	radio frequency
RNG	range

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RTN	return
SEL	select
SRA(s)	shop replaceable assembly(ies)
STBY	standby
TTL	transistor-transistor logic
TTG	time-to-go
TV	television
U, UPR	upper
V	volt
(V)	validity
VOM	volt-ohmmeter
VTVM	vacuum-tube voltmeter
WDC	weapon delivery computer
WRA(s)	weapons replaceable assembly(ies)
ΔB	weapon burst height
$\pm$	plus or minus
$\leq$	less than or equal to

6.7 Associated Equipment -

	Equipment	Designation
(1)	A-4M aircraft control panel	ICD 7941503
(2)	AV-8A aircraft control panel	DAC-A 2598
(3)	Weapon Delivery Computer (WDC)	MIL-D-85057
(4)	Dual Mode Tracker (DMT)	MIL-D-85056

6.8 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liaibility whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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6.9 This specification is under the cognizance of the Naval Air Systems Command (AIR-54932), Washington, D.C. 20361.

Procuring Activity
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